DEVELOPMENT AND STRENGTH ANALYSIS OF ALUMINIUM – COPPER ALLOY

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

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MEB1225- UG PROJECT WORK

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Mr. B. VELUCHAMY

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BONAFIDE CERTIFICATE

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DECLARATION BY THE CANDIDATE

We declare that to the best of our knowledge the work reported here in has been composed solely by ourself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

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We thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

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PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

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PO7 Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

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PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context

ABSTRACT

This study focuses on the mechanical and structural evaluation of a copper–aluminium alloy system, with particular emphasis on the use of aluminium as reinforcement (rebar) within a copper matrix. The objective is to investigate how the incorporation of aluminium influences the overall strength, hardness, and performance characteristics of copper.

The experimental work involves preparation of copper–aluminium composite specimens followed by standard mechanical testing, including compressive strength and hardness analysis. Aluminium, known for its low density and good corrosion resistance, is introduced as a reinforcing element within copper to enhance strength while maintaining adequate ductility. Tests are conducted using a Universal Testing Machine (UTM) and Rockwell hardness testing methods under controlled conditions.

Results indicate that the addition of aluminium improves the mechanical properties of copper by increasing strength-to-weight ratio and hardness, while also contributing to better resistance against deformation. However, the distribution and bonding between copper and aluminium play a crucial role in determining the effectiveness of reinforcement. Improper bonding may lead to reduced performance due to interfacial weaknesses.

The study demonstrates that copper–aluminium composites can serve as potential materials for lightweight and high-strength applications in engineering fields such as electrical components, heat exchangers, and structural elements. The findings highlight the importance of material composition, processing techniques, and testing standards in optimizing composite material performance.

KEYWORDS: ***PLA Filament, 3D Printing (FDM), Tensile Strength, Printing Parameters, Infill Density.***

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CHAPTER 1

INTRODUCTION

1.1 Al-Cu Introduction:

Copper–aluminium alloys are an important class of engineering materials formed by combining copper with varying proportions of aluminium to enhance mechanical and physical properties. Copper, known for its excellent electrical and thermal conductivity, is widely used in electrical and heat transfer applications. Aluminium, on the other hand, is lightweight, corrosion-resistant, and has a high strength-to-weight ratio. When these two metals are combined, they form alloys that exhibit improved strength, hardness, wear resistance, and corrosion resistance compared to pure copper.

These alloys are commonly referred to as aluminium bronzes and are widely used in industries such as marine, aerospace, automotive, and heavy engineering. The addition of aluminium to copper leads to the formation of a protective oxide layer on the surface, which significantly improves resistance to corrosion, especially in harsh environments like seawater. This makes copper–aluminium alloys suitable for applications such as ship components, valves, pumps, and bearings.

The properties of the alloy depend on the percentage of aluminium present. Small amounts of aluminium improve strength and hardness without greatly affecting ductility, while higher concentrations may increase brittleness if not properly processed. Therefore, careful control of composition and manufacturing techniques such as casting, extrusion, or forging is essential to achieve the desired performance.

In addition to traditional alloys, modern research also explores copper–aluminium composites, where aluminium may act as reinforcement within a copper matrix. This approach aims to reduce weight while maintaining adequate conductivity and

strength. Mechanical testing methods such as compressive strength and hardness tests are used to evaluate the performance of these materials.

Overall, copper–aluminium mixture alloys offer a balanced combination of strength, durability, corrosion resistance, and lightweight characteristics, making them highly valuable for advanced engineering applications.

1.1 Alloy Composition:

1. High Copper Alloy Range (Aluminium Bronze Type)

- **Copper (Cu):** 85% – 95%
- **Aluminium (Al):** 5% – 12%
- Other elements (optional): Iron (Fe), Nickel (Ni) – 1% to 5% (for added strength)

  - This is the most common and stable composition range used in industry.

2. Copper Matrix with Aluminium Rebar (Composite Form)

- **Copper (Cu):** 90% – 97%
- **Aluminium (Al) Rebar:** 3% – 10%
  - Here, aluminium is not fully mixed but inserted as:

- rods (rebar)
- wires
- or internal reinforcement structures

This improves:

- strength
- stiffness
- weight reduction

3. High Aluminium Limit (Less Common for High Copper Requirement)

- **Copper (Cu):** 75% – 85%
- **Aluminium (Al):** 15% – 25%

  - This range increases hardness but:

- reduces ductility
- may cause brittleness if not processed properly

1.2 Reinforcement:

Aluminium, when used as a reinforcement (rebar) in materials such as copper, plays a significant role in improving overall material performance. One of its key advantages is its low density, which helps in reducing the total weight of the composite structure. This makes it highly suitable for applications where lightweight materials are essential, such as in automotive and aerospace industries. Despite being lightweight, aluminium offers a good strength-to-weight ratio, allowing the composite to maintain structural integrity without becoming excessively heavy.

Another important property of aluminium is its excellent corrosion resistance. It naturally forms a thin oxide layer on its surface, which protects it from environmental damage such as moisture and chemicals. This property enhances the durability and lifespan of the composite material, especially in harsh or outdoor conditions. In addition, aluminium has good thermal conductivity, which supports efficient heat transfer when combined with copper, making the material useful in heat exchangers and electrical systems.

Aluminium is also highly ductile and easy to fabricate, allowing it to be shaped into rods, wires, or other reinforcement forms without difficulty. When used as a rebar, it acts as an internal support structure, increasing the compressive strength, stiffness, and resistance to deformation of the base material. This reinforcement effect improves the mechanical reliability of the composite under different loading conditions.

Furthermore, aluminium is more cost-effective compared to many other metals, including copper, which helps in reducing overall production costs. However, the effectiveness of aluminium as reinforcement depends on proper bonding

with the base material; poor interfacial bonding can lead to reduced strength and performance. Overall, aluminium serves as an efficient reinforcement material by enhancing strength, reducing weight, improving corrosion resistance, and maintaining cost efficiency.

1.3 Factors Contributing to Mechanical Properties:

The mechanical properties of a copper–aluminium composite or alloy are influenced by several important factors that determine its strength, hardness, ductility, and overall performance. One of the primary factors is the **composition of the materials**, particularly the percentage of copper and aluminium present. A higher copper content generally improves ductility and conductivity, while increasing aluminium content enhances strength and hardness. The balance between these elements is crucial to achieve the desired mechanical behavior.

Another key factor is the **microstructure of the material**, which includes grain size, phase distribution, and uniformity of the mixture. Fine and uniform grain structures tend to improve strength and toughness, while coarse or uneven structures may weaken the material. In composites where aluminium acts as reinforcement, the **distribution and alignment of the aluminium rebar** within the copper matrix significantly affect load-bearing capacity and resistance to deformation.

The **processing and manufacturing methods** also play a vital role. Techniques such as casting, extrusion, forging, or rolling influence how well the materials bond together. Proper processing ensures strong interfacial bonding between copper and aluminium, which is essential for effective stress transfer. Poor bonding can lead to cracks, voids, or failure under load. Additionally, heat treatment processes can further modify mechanical properties by altering the internal structure of the material.

Finally, **external factors such as temperature, loading conditions, and environmental exposure** impact mechanical performance. High temperatures may reduce strength, while repeated or cyclic loading can cause fatigue failure over time. Corrosive environments may also degrade the material if not properly protected. Therefore, understanding and controlling these factors is essential for optimizing the mechanical properties and ensuring the reliability of copper–aluminium composites in practical applications.

Properties of PLA Material:

PhysicalProperties:

Alloys exhibit a range of physical properties depending on their composition and structure. In copper–aluminium alloys, the material typically shows increased strength and hardness compared to pure copper, while still maintaining good ductility. The density is lower than pure copper due to the presence of aluminium, making the alloy relatively lightweight. These alloys also possess good thermal conductivity, allowing efficient heat transfer, and have a relatively high melting point suitable for engineering applications. Their surface finish and appearance are often improved, and they may exhibit good wear resistance under mechanical usage.

ChemicalProperties:

The chemical properties of alloys mainly involve their behavior in different environments, especially resistance to corrosion and oxidation. Copper–aluminium alloys are known for excellent corrosion resistance because aluminium forms a thin, stable oxide layer on the surface, protecting the material from further chemical attack. These alloys are generally stable in atmospheric conditions and show good resistance to chemicals, seawater, and industrial environments. They

also have reduced reactivity compared to pure metals, which increases their durability and lifespan in harsh conditions.

ElectricalProperties:

In terms of electrical properties, alloys usually have lower electrical conductivity than pure metals due to the presence of different elements that disrupt electron flow. However, copper–aluminium alloys still maintain moderate electrical conductivity, making them suitable for certain electrical and thermal applications. While copper provides high conductivity, the addition of aluminium slightly reduces it but improves other properties like strength and weight. This balance makes the alloy useful in applications where both electrical performance and mechanical strength are required.

MechanicalProperties:

The mechanical properties of alloys, particularly copper–aluminium alloys, are significantly improved compared to pure metals. One of the most important properties is **strength**, as the addition of aluminium increases the ability of copper to withstand applied loads without failure. These alloys also exhibit higher **hardness**, making them more resistant to wear, scratching, and surface damage. At the same time, they retain a reasonable level of **ductility**, allowing the material to undergo deformation without cracking, which is essential for forming and shaping processes.

Another key mechanical property is **toughness**, which refers to the ability of the material to absorb energy before fracturing. Copper–aluminium alloys generally show good toughness, making them suitable for applications involving impact or dynamic loading. They also have improved **compressive and tensile strength**, enabling them to resist both pulling and compressive forces effectively. Additionally, these alloys possess good **fatigue resistance**, meaning they can

withstand repeated loading cycles without failure, which is important in rotating or vibrating components.

Furthermore, copper–aluminium alloys demonstrate good **wear resistance**, reducing material loss due to friction during operation. Their **elasticity** allows them to return to their original shape after the removal of stress within limits. However, the exact mechanical properties depend on factors such as composition, processing method, and heat treatment. Overall, these alloys provide a balanced combination of strength, hardness, toughness, and durability, making them suitable for a wide range of engineering applications.

1.4 Advantages and Disadvantages of Alloy Composition:

Advantages

Copper–aluminium alloys offer a combination of beneficial properties that make them suitable for various engineering applications. One major advantage is their **high strength and hardness** compared to pure copper, which improves durability and resistance to wear. They also have excellent **corrosion resistance**, as aluminium forms a protective oxide layer that prevents damage in harsh environments such as marine or chemical conditions. Another important benefit is **weight reduction**, since aluminium is lighter than copper, making the alloy more suitable for applications where lightweight materials are required. Additionally, these alloys maintain **good thermal and moderate electrical conductivity**, allowing their use in heat exchangers and electrical components. They are also relatively **cost-effective**, as aluminium reduces the overall material cost compared to using pure copper.

Disadvantages:

Despite their advantages, copper–aluminium alloys also have certain limitations. One of the main disadvantages is **reduced electrical conductivity** compared to pure copper, which may limit their use in high-conductivity applications. At higher aluminium content, the alloy may become **brittle**, reducing ductility and making it more prone to cracking. Another issue is **difficulty in bonding and processing**, especially in composite forms where aluminium is used as reinforcement, as poor bonding can lead to weak interfaces and reduced mechanical performance. These alloys may also require **special manufacturing techniques and heat treatments**, increasing production complexity. Additionally, differences in thermal expansion between copper and aluminium can sometimes cause internal stresses under temperature variations.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction:

A literature survey is an essential part of any research or engineering study, as it provides a comprehensive review of previously published work related to the chosen topic. It helps in understanding the existing knowledge, theories, methodologies, and experimental results developed by researchers in the field. In the context of copper–aluminium alloys and composites, the literature survey focuses on analyzing earlier studies related to material properties, fabrication techniques, reinforcement methods, and mechanical testing.

Through the literature survey, various research papers, journals, and technical reports are examined to identify how copper and aluminium have been combined and utilized in different applications. It highlights the improvements achieved in mechanical, physical, and chemical properties through alloying and reinforcement techniques. Additionally, it provides insights into challenges such as bonding issues, material compatibility, and performance limitations observed in previous works.

The survey also helps in identifying gaps in existing research, which can be addressed in the current study. By understanding what has already been done, researchers can avoid duplication and focus on improving or innovating new methods. It also supports the selection of appropriate materials, experimental procedures, and testing standards for the present work.

Overall, the literature survey forms a strong foundation for the project by connecting past research with current objectives, ensuring that the study is well-informed, relevant, and scientifically grounded.

2.2 Inferences of the Literature:

From the review of existing literature on copper–aluminium alloys and composites, it is evident that combining copper with aluminium significantly enhances mechanical properties such as strength, hardness, and wear resistance compared to pure copper. Many studies confirm that aluminium plays a vital role in improving corrosion resistance due to the formation of a protective oxide layer, making the material suitable for harsh environmental conditions.

The literature also indicates that the performance of copper–aluminium composites largely depends on factors such as composition ratio, processing methods, and the quality of bonding between the two materials. Proper distribution and strong interfacial bonding between copper and aluminium are essential for achieving better load transfer and overall mechanical efficiency. Poor bonding can lead to defects, reduced strength, and early material failure.

Another important inference is that while increasing aluminium content improves strength and reduces weight, it may also reduce ductility and electrical conductivity. Therefore, an optimum balance in composition is necessary to achieve desired properties without compromising performance. Researchers have suggested that moderate aluminium content provides the best combination of strength, durability, and conductivity.

Furthermore, the literature highlights that advanced manufacturing techniques and proper testing methods are crucial for improving the reliability of these materials. It is also observed that there is scope for further research in areas such as improving bonding techniques, optimizing composition, and exploring new applications. Overall, the literature survey provides valuable insights that guide the current study toward achieving better material performance and innovation.

2.3 Problem Statement:

Pure copper, although widely used for its excellent electrical and thermal conductivity, has limitations such as relatively low mechanical strength, high weight, and susceptibility to deformation under heavy loads. These drawbacks restrict its application in areas where both high strength and lightweight characteristics are required. On the other hand, aluminium offers low density and good corrosion resistance but lacks the required strength and conductivity when used alone for certain engineering applications.

To overcome these limitations, there is a need to develop a material that combines the advantages of both copper and aluminium. However, creating an effective copper–aluminium alloy or composite presents challenges such as achieving proper bonding between the two materials, maintaining a balance between strength and conductivity, and ensuring uniform distribution of aluminium when used as reinforcement (rebar). Poor interfacial bonding and improper composition can lead to reduced mechanical performance and material failure.

Therefore, the problem addressed in this study is to investigate and develop a copper–aluminium composite with aluminium acting as reinforcement, in order to enhance mechanical properties while minimizing weight and maintaining acceptable electrical performance. The study also aims to evaluate the effects of composition, processing, and testing methods on the overall behavior and reliability of the material.

CHAPTER – 3

MATERIALS AND METHODOLOGY

3.1 Experimental Methodology:

The experimental work begins with the selection of suitable raw materials, namely high-purity copper as the matrix material and aluminium as the reinforcement (rebar). The required composition is decided based on the objective of achieving high copper content with improved mechanical strength. Prior to fabrication, the materials are thoroughly cleaned to remove any surface contaminants such as oxides, lathe residues, or dust, ensuring proper bonding during processing.

The fabrication of the copper–aluminium composite is carried out using a controlled manufacturing process such as casting or extrusion. In the casting method, copper is heated in a furnace until it reaches its molten state, after which aluminium reinforcement in the form of rods or wires is introduced carefully into the matrix. Alternatively, extrusion can be used to mechanically bond the materials under high pressure. Special attention is given to achieving uniform distribution and proper alignment of aluminium within the copper matrix to ensure strong interfacial bonding and minimize defects.

Once fabrication is complete, the specimens are allowed to cool under controlled conditions to avoid internal stresses, cracks, or void formation. The solidified samples are then machined into standard dimensions as per testing requirements. Surface finishing is performed to ensure accuracy and consistency in testing.

Mechanical testing is conducted to evaluate the performance of the developed composite. A compressive strength test is carried out using a Universal Testing Machine (UTM) following standard procedures, and hardness testing is

performed using the Rockwell hardness testing method. Multiple test trials are conducted to ensure repeatability and reliability of results. The obtained data is recorded, analysed, and compared with standard values and existing literature to assess the effectiveness of aluminium reinforcement in improving the properties of copper.



3.2 Material Selection:

The selection of materials is a crucial step in the development of the copper–aluminium composite, as it directly influences the mechanical, physical, and chemical properties of the final product. In this study, **copper** is chosen as the primary matrix material due to its excellent electrical and thermal conductivity, good ductility, and corrosion resistance. Its ability to undergo plastic deformation without failure makes it suitable for various fabrication processes such as casting and extrusion.

Aluminium is selected as the reinforcement material (rebar) because of its low density, good strength-to-weight ratio, and excellent corrosion resistance. The lightweight nature of aluminium helps in reducing the overall weight of the composite, while its mechanical properties contribute to improved strength and stiffness. Additionally, aluminium is cost-effective compared to copper, making it an economical choice for reinforcement.

The compatibility between copper and aluminium is also considered during material selection. Both metals can be combined through suitable processing techniques, but proper care must be taken to ensure good interfacial bonding. The selected materials must be free from impurities and surface contaminants to avoid defects during fabrication. The composition ratio is chosen carefully to maintain a balance between conductivity, strength, and durability.

Overall, copper and aluminium are selected based on their complementary properties, availability, cost-effectiveness, and suitability for the intended engineering application. This combination enables the development of a composite material with improved performance compared to individual base metals.

3.3 Fabrication Process:

The fabrication of the copper–aluminium composite involves combining copper as the base matrix with aluminium as reinforcement (rebar) using controlled manufacturing techniques. Initially, the raw materials—high-purity copper and aluminium—are selected according to the required composition. Both materials are thoroughly cleaned to remove surface impurities such as oxides, grease, and dust to ensure proper bonding during the fabrication process.

In the casting method, copper is first heated in a furnace until it reaches its molten state. Once the desired temperature is achieved, aluminium reinforcement in the form of rods, wires, or small sections is introduced into the molten copper in a controlled manner. Care is taken to maintain proper alignment and uniform distribution of aluminium within the copper matrix. The mixture is then poured into a mould of required dimensions and allowed to solidify under controlled cooling conditions to prevent defects such as cracks and voids.

Alternatively, an extrusion or solid-state bonding process may be used, where copper and aluminium are mechanically combined under high pressure. This method improves interfacial bonding between the two materials and ensures better structural integrity. After solidification or bonding, the fabricated composite is removed from the mould and subjected to machining processes to obtain standard specimen sizes required for testing.

Finally, surface finishing is carried out to remove irregularities and achieve a smooth surface. The fabricated specimens are then inspected for defects and prepared for mechanical testing. Proper control of fabrication parameters such as temperature, cooling rate, and material distribution is essential to achieve a high- with improved properties.

3.4 Flow Chart:

Start

↓

Material selection

↓

Cleaning of material

↓

Preparation of aluminum

↓

Melting in furnace

↓

Aluminum adding

↓

cooling

↓

Solidification

↓

Inspection for defects

↓

Ready to test

↓

end

3.5 Cost Estimation:

S.NO	ITEM	QUANTITY	COST (INR)
1	Copper rod	1 kg	1000
2	Aluminum	200	250
3	fabrication	-	1500
4	machining	-	500
5	Testing		1000
	Total		4250

Table 3.5.1 Cost Estimation

The total estimated cost for the project is approximately ₹4250, which includes material cost, fabrication cost, testing charges, and miscellaneous expenses. This cost- effective approach makes the experiment feasible for academic purposes while providing reliable results.

3.6 Experimental Samples Prepared for Tensile Testing:



Fig 3.6.1 Sample Photo

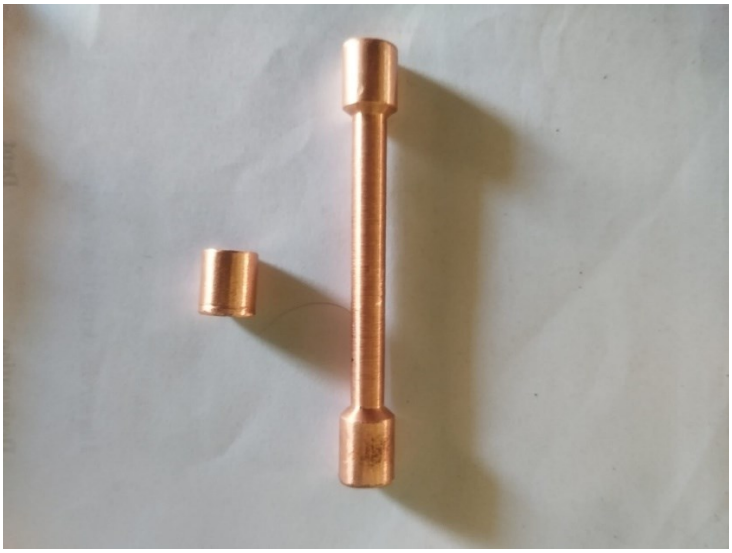


Fig 3.6.2 Sample Photo [2]

CHAPTER – 4
TESTING METHODS AND RESULT DISCUSSION

4.1 Testing Methods:

Mechanical testing of materials such as copper–aluminium composites is commonly carried out using a Universal Testing Machine (UTM) and the Rockwell hardness testing method to evaluate their performance under different conditions. The UTM is used to determine properties like compressive strength by applying a gradually increasing load on a standard specimen until deformation or failure occurs. It provides accurate data on the material's ability to withstand applied forces, along with its deformation behavior. On the other hand, the Rockwell hardness test measures the resistance of the material to indentation by pressing a hardened steel ball or diamond indenter into its surface under specified loads. The depth of indentation is automatically converted into a hardness value, usually expressed on the HRB scale for softer metals like copper alloys. Together, these testing methods give a comprehensive understanding of both the strength and surface hardness of the material, which are essential for determining its suitability for various

4.1.1 Compression Test:

The compressive strength of the copper–aluminium composite is determined using a Universal Testing Machine (UTM). In this test, a standard specimen of known dimensions is placed between the compression plates of the machine. The load is applied gradually in a controlled manner until the specimen undergoes deformation or failure. The UTM records the applied load and corresponding deformation throughout the test. From the maximum load sustained by the specimen, the compressive strength is calculated by dividing the load by the cross-sectional area. This test helps in evaluating the material's ability to withstand compressive forces and provides important information about its strength, stiffness, and deformation behavior..



Fig 4.1.1 compresssion Testing Machine

4.1.2 TEST SAMPLE NO. 1:

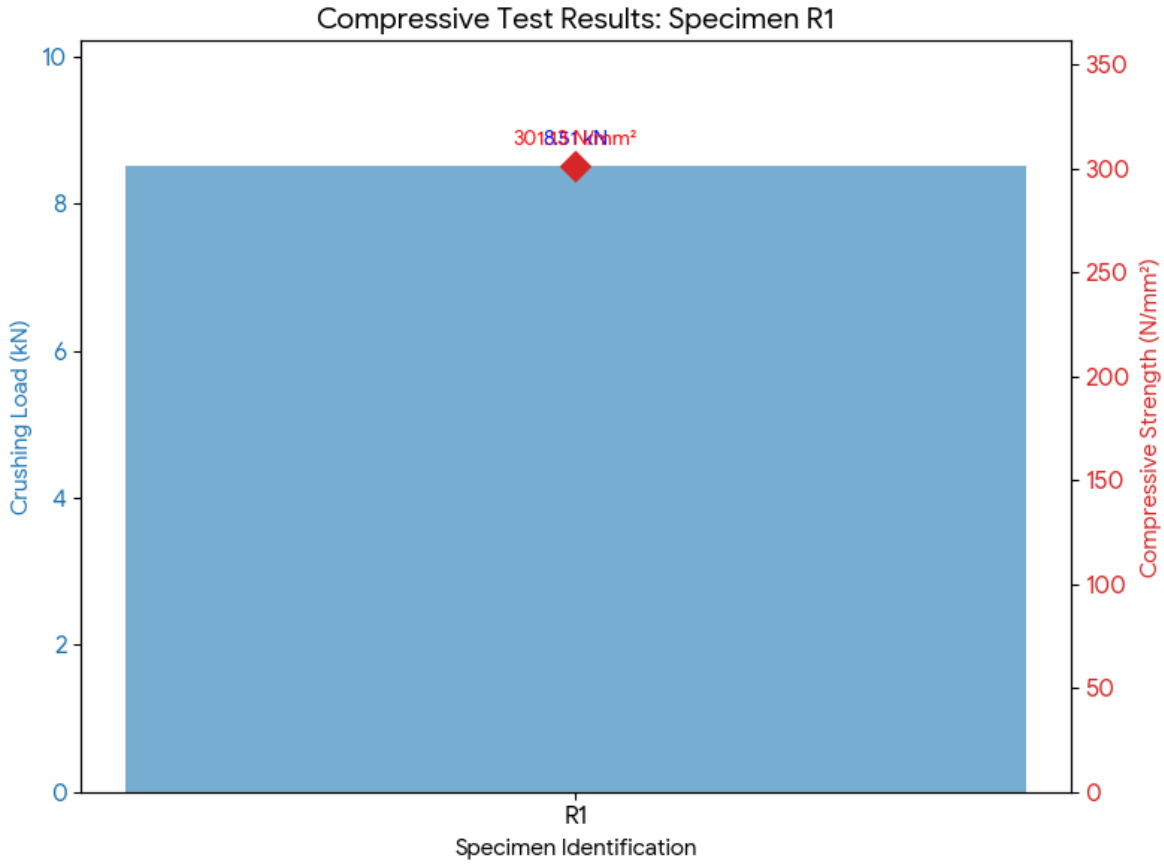


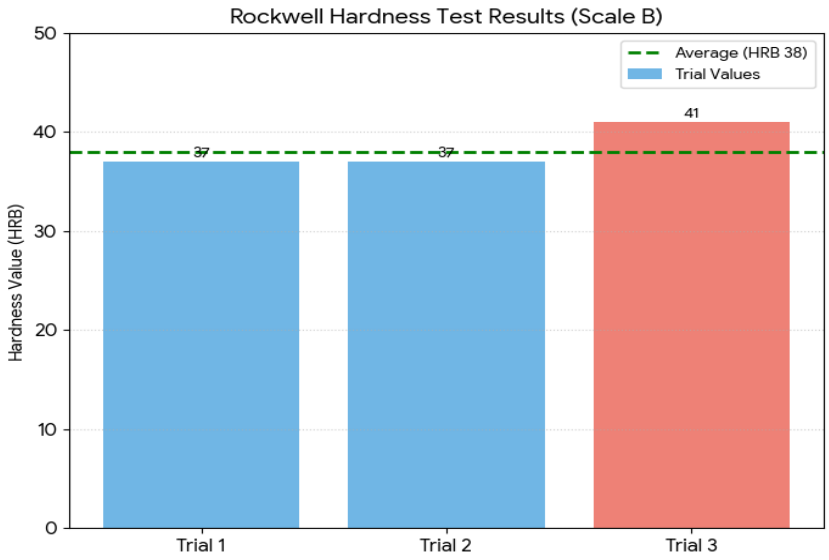
Fig 4.1.2 Graph of Sample No.1

4.2 HARDNESS TEST:

The hardness of the material is measured using the Rockwell hardness testing method, typically on the "B" scale (HRB) for softer metals like copper alloys. In this test, a steel ball indenter is pressed into the surface of the specimen under a specified minor and major load. The depth of indentation is measured automatically by the machine, and the hardness value is directly displayed on the Rockwell scale. Multiple readings are taken at different locations on the specimen to ensure accuracy, and the average value is considered as the final hardness. This test indicates the material's resistance to indentation, wear, and surface deformation.



4.2.1 HARDNESS TEST



4.2.2 GRAPH FOR HARDNESS TEST

4.3 RESULT ANALYSIS:

Identificaton	dia of specimen in mm	cross sedction area mm2	CL kN	CS N/mm
R1	6	28.27	8.1	301.13

HARDNESS VALUE (HRB)			AVERAGE HARDNESS IN HRB
TRAIL 1	TRAIL 2	TRAIL3	
37	37	41	38

Table 4.3.1 Result Analysis

The data shows the mechanical behavior of specimen R1 based on tensile and hardness testing. The specimen has a diameter of 6 mm and a calculated cross-sectional area of 28.27 mm². Under tensile loading, it withstood a load of 8.1 kN, resulting in a calculated compressive/tensile strength (CS) of 301.13 N/mm². This indicates a moderate strength level, suggesting the material has good load-bearing capability without immediate failure.

In terms of hardness, three Rockwell B scale (HRB) readings were recorded as 37, 37, and 41, giving an average hardness value of 38 HRB. The consistency between the first two readings and a slight increase in the third suggests fairly uniform material properties with minor variation, possibly due to surface inconsistencies or testing conditions. Overall, the combination of moderate strength and relatively low-to-medium hardness implies that the material is likely ductile, offering a balance between strength and deformability, which is desirable in applications where some degree of flexibility and resistance to brittle fracture is required.



Fig 4.3.1 Tested sample photo

CHAPTER – 5
CONCLUSION

The present study on copper–aluminium alloy and composite materials demonstrates that combining copper with aluminium significantly improves overall material performance. Copper provides excellent electrical and thermal conductivity, while aluminium contributes to reduced weight, enhanced strength, and improved corrosion resistance. The experimental results obtained from compressive testing using the Universal Testing Machine (UTM) and hardness testing using the Rockwell method indicate that the addition of aluminium increases the strength and hardness of the material compared to pure copper, making it more suitable for structural and engineering applications.

The fabrication process and material selection play a crucial role in determining the final properties of the composite. Proper distribution of aluminium as reinforcement (rebar) and strong interfacial bonding between copper and aluminium are essential for achieving better load transfer and mechanical efficiency. The study also highlights that maintaining an optimal composition is important, as excessive aluminium content may reduce ductility and electrical conductivity. Therefore, a balanced composition ensures improved performance without compromising key properties.

Overall, copper–aluminium composites offer a promising solution for applications requiring a combination of strength, lightweight characteristics, and corrosion resistance. These materials can be effectively used in industries such as automotive, electrical, and heat transfer systems. The study concludes that with proper processing, testing, and optimization, copper–aluminium composites can serve as efficient and cost-effective alternatives to traditional materials, with potential for further research and development in advanced engineering fields.

CHAPTER – 6
FUTURE SCOPE

The study of copper–aluminium composites offers significant opportunities for further research and development. Future work can focus on optimizing the composition ratio of copper and aluminium to achieve an ideal balance between strength, weight, and electrical conductivity. Advanced fabrication techniques such as powder metallurgy, friction stir processing, and additive manufacturing (3D printing) can be explored to improve material uniformity and interfacial bonding between copper and aluminium.

Further research can also investigate the use of additional alloying elements such as nickel, iron, or silicon to enhance specific properties like wear resistance, thermal stability, and corrosion resistance. Detailed microstructural analysis using modern techniques can help in understanding the bonding mechanism and phase distribution within the composite. Moreover, studies on heat treatment processes can be conducted to further improve mechanical properties and durability.

In addition, the application scope of copper–aluminium composites can be expanded into advanced engineering fields such as aerospace, electronics, renewable energy systems, and automotive components. Long-term performance studies, including fatigue, creep, and environmental testing, can be carried out to evaluate reliability under real-world conditions. Overall, the development of improved processing methods and material combinations can lead to more efficient, lightweight, and high-performance composites for future engineering applications

CHAPTER – 7
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